

Trexquant — GP Alphas

21 qualifying alphas from 828,000 evaluated expressions (30 seeds x 600 population x 45 generations). Trained 2011-2017, validated on held-out 2018-2020. Generated from gp_results_topk.csv.

How to use this

Ordered by held-out valid_ir, not by in-sample fitness. Across the full result set the rank correlation between in-sample fitness and out-of-sample IR was **-0.70** — negative, so the in-sample ranking is actively misleading. Work down this list in order.

Every alpha here clears all four platform gates: daily IR > 0.07, turnover < 0.5, stock count > 160, and positive held-out IR. Maximum pairwise PnL correlation across the whole set is 0.49 and the mean is 0.10, so none should breach the 50% gate against another.

Two things that will silently break an alpha. (1) The - 1 after every cs_rank is deliberate: Trexsim's cs_rank returns [1,2] while the GP worked in [0,1]. Removing it changes the alpha. (2) Give slow mode a generous lookback — nested operators compound their windows, so an expression with w60 inside w20 needs far more history than 60 days. If fast and slow disagree, raise the lookback first.

Copying from a PDF can insert stray line breaks into long expressions. GP_Alphas.txt next to this file has the identical strings as plain text — prefer it for pasting.

Summary

#	valid IR	daily IR	TVR	net IR	stocks	max corr
1	1.89	0.1019	0.102	1.31	325	0.40
2	1.39	0.0868	0.058	1.12	317	0.42
3	1.30	0.1306	0.206	1.19	325	0.37
4	1.20	0.1240	0.176	1.25	322	0.34
5	1.20	0.1423	0.199	1.50	311	0.32
6	1.19	0.0881	0.045	1.17	326	0.12
7	1.10	0.0887	0.062	1.11	318	0.11
8	1.09	0.1317	0.190	1.27	328	0.38
9	1.01	0.0750	0.066	0.92	330	0.49
10	0.71	0.0825	0.051	1.12	317	0.31
11	0.69	0.0871	0.057	1.14	324	0.21
12	0.67	0.0907	0.050	1.22	323	0.22
13	0.65	0.0985	0.053	1.41	306	0.27
14	0.58	0.1015	0.023	1.52	303	0.07
15	0.56	0.1592	0.204	1.64	322	0.21
16	0.51	0.1292	0.196	1.20	330	0.49
17	0.51	0.0995	0.033	1.55	307	0.27
18	0.47	0.0821	0.055	1.07	319	0.15
19	0.39	0.0915	0.063	1.17	311	0.33
20	0.13	0.0830	0.073	0.99	316	0.13
21	0.04	0.1077	0.077	1.38	323	0.10

Turnover is worth watching. The tutorial's own worked examples run 0.21-1.58; anything far below that barely trades, and the one alpha tested live so far (TVR 0.045) underperformed. Prefer the higher-TVR entries first.

Alpha 1 — csv rank 22, seed 19

held-out IR **1.89** | daily IR **0.1019** (gate 0.07) | TVR **0.102** (limit 0.5) | net IR 1.31 | stocks 325 (min 160) | drawdown 1.7% | max corr 0.40

Paste into Trexsim:

```
alpha = (((at_signlog(ts_mean_exp(-(ts_corr_binary(ret1, ret1_spx, 60)), 60, 0.5)) - (((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))) - (cs_rank(ts_mean(close * volume, 60)) - 1)))) - (ts_std((ts_std(((close * volume) / at_zero2nan(ts_mean(close * volume, 60))), 60) * cs_zscore(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))))) , 60) * cs_zscore(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))))) - cs_zscore(cs_zscore((cs_rank(ts_mean(close * volume, 60)) - 1))))
```

GP form: sub(sub(sub(signlog(ts_decay(neg(beta60), w60)), neg(sub(stoch, size))), mul(ts_std(mul(ts_std(dv_adv, w60), cs_zscore(stoch)), w60), cs_zscore(stoch))), cs_zscore(cs_zscore(size)))

Alpha 2 — csv rank 17, seed 12

held-out IR **1.39** | daily IR **0.0868** (gate 0.07) | TVR **0.058** (limit 0.5) | net IR 1.12 | stocks 317 (min 160) | drawdown 2.0% | max corr 0.42

Paste into Trexsim:

```
alpha = ts_mean_exp((ts_median((cs_rank(ts_std(ts_max(at_signlog(((close * volume) / at_zero2nan(ts_mean(close * volume, 60))) + (close / at_zero2nan(ts_mean(close, 60))))) , 60), 60)) - 1), 20) - at_signsqrt(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))))) , 10, 0.5)
```

GP form: ts_decay(sub(ts_median(cs_rank(ts_std(ts_max(signlog(add(dv_adv, c_ma60)), w60), w60)), w_id(w_id(w_id(w_id(w_id(w20))))), signsqrt(stoch)), w_id(w_id(w10)))

Alpha 3 — csv rank 29, seed 8

held-out IR **1.30** | daily IR **0.1306** (gate 0.07) | TVR **0.206** (limit 0.5) | net IR 1.19 | stocks 325 (min 160) | drawdown 1.2% | max corr 0.37

Paste into Trexsim:

```
alpha = ts_mean_exp((((-(ts_zscore(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))), 5)) - ((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20)))) - ts_corr_binary(ts_corr_binary(ret1, ret1_spx, 60), ts_corr_binary(ts_corr_binary(ret1, ret1_spx, 60), ((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1))), 20), 20)) - ts_corr_binary(ts_corr_binary(ret1, ret1_spx, 60), ((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1))), 20)), 20, 0.5)
```

GP form: ts_decay(sub(sub(sub(neg(ts_zscore(stoch, w_id(w_id(w5)))), stoch), ts_corr(beta60, ts_corr(beta60, gap, w20), w20)), ts_corr(beta60, gap, w20)), w_id(w_id(w20)))

Alpha 4 — csv rank 26, seed 11

held-out IR **1.20** | daily IR **0.1240** (gate 0.07) | TVR **0.176** (limit 0.5) | net IR 1.25 | stocks 322 (min 160) | drawdown 1.9% | max corr 0.34

Paste into Trexsim:

```
alpha = ts_sum(-(ts_zscore(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))), 10)) - (((ts_median(cs_zscore((volume / at_zero2nan(ts_mean(volume, 20))), 10) - ts_mean(ret20, 10)) + ts_skew(((volume / at_zero2nan(ts_mean(volume, 20))) - ts_delay((volume / at_zero2nan(ts_mean(volume, 20))), 10)) - ts_delay(((volume / at_zero2nan(ts_mean(volume, 20))) - ts_delay((volume / at_zero2nan(ts_mean(volume, 20))), 10))), 10)), 20)) - ts_mean(-(ts_median(ts_skew(((close - open) / at_zero2nan(open)), 60), 60)) - ts_mean(ret20, 10)), 10)), 10)
```

GP form: ts_sum(sub(neg(ts_zscore(stoch, w_id(w_id(w_id(w10))))), sub(add(sub(ts_median(cs_zscore(v_ma20, w_id(w_id(w10))), ts_mean(ret20, w_id(w_id(w10))), ts_skew(ts_delta(ts_delta(v_ma20, w10), w10), w_id

Alpha 5 — csv rank 21, seed 28

held-out IR **1.20** | daily IR **0.1423** (gate 0.07) | TVR **0.199** (limit 0.5) | net IR 1.50 | stocks 311 (min 160) | drawdown 1.9% | max corr 0.32

Paste into Trexsim:

```
alpha = (-(ts_mean_exp(ts_zscore(((close / at_zero2nan(ts_mean(close, 20))) + ts_max((close / at_zero2nan(ts_mean(close, 60))), 60)), 10), 20, 0.5)) - ts_mean_exp((cs_rank((ts_zscore(ts_std(((close / at_zero2nan(ts_mean(close, 60))) - ts_delay((close / at_zero2nan(ts_mean(close, 60))), 60)), 20), 60) + ts_mean_exp(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))), 20, 0.5))) - 1), 10, 0.5)))
```

GP form: sub(neg(ts_decay(ts_zscore(add(c_ma20, ts_max(c_ma60, w60))), w_id(w10)), w20)), ts_decay(cs_rank(add(ts_zscore(ts_std(ts_delta(c_ma60, w60), w_id(w_id(w20))), w_id(w60))), ts_decay(stoch, w20))), w10))

Alpha 6 — csv rank 12, seed 14

held-out IR **1.19** | daily IR **0.0881** (gate 0.07) | TVR **0.045** (limit 0.5) | net IR 1.17 | stocks 326 (min 160) | drawdown 1.2% | max corr 0.12

Paste into Trexsim:

```
alpha = -(ts_sum(ts_std(ts_min(ts_corr_binary((ret20 - (cs_rank(ts_mean(close * volume, 60)) - 1)), ((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))), 5), 10), 60), 20))
```

GP form: neg(ts_sum(ts_std(ts_min(ts_corr(sub(ret20, size), stoch, w5), w_id(w_id(w_id(w_id(w10))))), w_id(w_id(w60))), w20))

Alpha 7 — csv rank 18, seed 3

held-out IR **1.10** | daily IR **0.0887** (gate 0.07) | TVR **0.062** (limit 0.5) | net IR 1.11 | stocks 318 (min 160) | drawdown 1.7% | max corr 0.11

Paste into Trexsim:

```
alpha = ts_sum(-(at_signlog(ts_corr_binary(ret20, (ts_corr_binary(ts_skew(ts_corr_binary(ret1, ret1_spx, 60), 10), at_signsqrt(((close * volume) / at_zero2nan(ts_mean(close * volume, 60)))), 60) - ((ts_delay((close / at_zero2nan(ts_mean(close, 20))), 20)) - ts_mean_exp(at_signsqrt(((close * volume) / at_zero2nan(ts_mean(close * volume, 60))), 20, 0.5))), 60))), 10)
```

GP form: ts_sum(neg(signlog(ts_corr(ret20, sub(ts_corr(ts_skew(beta60, w10), signsqrt(dv_adv), w_id(w60))), sub(neg(ts_delay(c_ma20, w20)), ts_decay(signsqrt(dv_adv), w20))), w_id(w_id(w60)))), w_id(w_id(w_id(w_id(w10))))), w20))

Alpha 8 — csv rank 25, seed 24

held-out IR **1.09** | daily IR **0.1317** (gate 0.07) | TVR **0.190** (limit 0.5) | net IR 1.27 | stocks 328 (min 160) | drawdown 1.7% | max corr 0.38

Paste into Trexsim:

```
alpha = (ts_mean((ts_zscore(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20)))) / at_zero2nan(ts_mean((cs_rank(ts_std(ret1, 20)) - 1), 20))) / at_zero2nan(ts_mean(ts_mean(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))) + (cs_rank(ts_mean(close * volume, 60)) - 1)), 10), 20))), 10) - (cs_rank(ts_mean(close * volume, 60)) - 1), 10) - (cs_rank(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))) + ((ts_skew(ts_median(((close - open) / at_zero2nan(open)), 20), 60) + ((cs_rank(ts_mean(close * volume, 60)) - 1) + ret5)) + ts_min(ts_mean(ts_mean((cs_rank(ts_mean(close * volume, 60)) - 1), 10), 20), 20, 20)))) - 1))
```

GP form: sub(ts_mean(sub(ts_zscore(div(div(neg(stoch), ts_mean(cs_rank(vol20, w20))), ts_mean(ts_mean(add(stoch, size), w_id(w10))), w20))), w10), size), w10), cs_rank(add(stoch, add(add(ts_skew(ts_median(co, w2

Alpha 9 — csv rank 27, seed 29

held-out IR **1.01** | daily IR **0.0750** (gate 0.07) | TVR **0.066** (limit 0.5) | net IR 0.92 | stocks 330 (min 160) | drawdown 2.6% | max corr 0.49

Paste into Trexsim:

```
alpha = (((-(ts_corr_binary(ret20, (cs_rank(ts_mean(close * volume, 60)) - 1), 60)) - (cs_rank(ts_mean(close * volume, 60)) - 1)) - ts_sum((close / at_zero2nan(ts_mean(close, 20))), 5)) - ts_max((cs_rank(ts_mean(close * volume, 60)) - 1), 5))
```

GP form: sub(sub(sub(neg(ts_corr(ret20, size, w_id(w60))), size), ts_sum(c_ma20, w5)), ts_max(size, w_id(w5)))

Alpha 10 — csv rank 16, seed 12

held-out IR **0.71** | daily IR **0.0825** (gate 0.07) | TVR **0.051** (limit 0.5) | net IR 1.12 | stocks 317 (min 160) | drawdown 1.7% | max corr 0.31

Paste into Trexsim:

```
alpha = ts_mean_exp((ts_median((cs_rank(ts_std(ts_max((ts_mean_exp(((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1))), 20, 0.5) / at_zero2nan(ts_sum(((close * volume) / at_zero2nan(ts_mean(close * volume, 60))), 60))), 60, 60)) - 1), 5) - at_signsqrt(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))))), 20, 0.5)
```

GP form: ts_decay(sub(ts_median(cs_rank(ts_std(ts_max(div(ts_decay(gap, w20), ts_sum(dv_adv, w60)), w60), w60)), w_id(w_id(w_id(w_id(w5))))), signsqrt(stoch)), w_id(w_id(w20)))

Alpha 11 — csv rank 14, seed 26

held-out IR **0.69** | daily IR **0.0871** (gate 0.07) | TVR **0.057** (limit 0.5) | net IR 1.14 | stocks 324 (min 160) | drawdown 1.7% | max corr 0.21

Paste into Trexsim:

```
alpha = (ts_max(ts_mean(ts_max(ts_corr_binary((ts_mean(ts_std(ret1, 20), 20) - ts_zscore(ts_std(ret1, 20), 10)), ret20, 60), 10), 20), 5) - (close / at_zero2nan(ts_mean(close, 20))))
```

GP form: sub(ts_max(ts_mean(ts_max(ts_corr(sub(ts_mean(vol20, w20), ts_zscore(vol20, w_id(w10))), ret20, w60), w10), w_id(w20)), w_id(w_id(w_id(w5)))), c_ma20)

Alpha 12 — csv rank 11, seed 27

held-out IR **0.67** | daily IR **0.0907** (gate 0.07) | TVR **0.050** (limit 0.5) | net IR 1.22 | stocks 323 (min 160) | drawdown 1.2% | max corr 0.22

Paste into Trexsim:

```
alpha = (((cs_rank(ts_mean(close * volume, 60)) - 1) + ts_mean_exp(-((cs_rank(ts_mean(close * volume, 60)) - 1)) + ts_mean_exp((((cs_rank(ts_mean(close * volume, 60)) - 1) - ((close - open) / at_zero2nan(open))) + (cs_rank(ts_mean(close * volume, 60)) - 1)) + (((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))) * ((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20)))) + (cs_rank(ts_mean(close * volume, 60)) - 1))), 10, 0.5)), 60, 0.5)) * ts_min((ts_skew(ts_mean_exp((((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))) + (cs_rank(ts_mean(close * volume, 60)) - 1))), 60, 0.5), 5) * ts_skew(-(cs_rank(ts_mean(close * volume, 60)) - 1)), 5)), 60))
```

GP form: mul(add(size, ts_decay(add(neg(size), ts_decay(add(add(sub(size, co), size), add(mul(stoch, stoch), size)), w_id(w_id(w10)))), w_id(w60))), ts_min(mul(ts_skew(ts_decay(add(stoch, size), w60), w5), ts_

Alpha 13 — csv rank 10, seed 22

held-out IR **0.65** | daily IR **0.0985** (gate 0.07) | TVR **0.053** (limit 0.5) | net IR 1.41 | stocks 306 (min 160) | drawdown 3.3% | max corr 0.27

Paste into Trexsim:

```
alpha = ts_std(ts_median((((close / at_zero2nan(ts_mean(close, 20))) - ts_delay((close / at_zero2nan(ts_mean(close, 20))), 10)) + ((ts_min(ts_mean_exp((ts_corr_binary(ret1, ret1_spx, 60) - ts_delay(ts_corr_binary(ret1, ret1_spx, 60), 10))), 60, 0.5), 60) - ts_min(-(close / at_zero2nan(ts_mean(close, 20)))), 60)) + (at_signlog(ts_min((ts_corr_binary(ret1, ret1_spx, 60) - ts_delay(ts_corr_binary(ret1, ret1_spx, 60), 10))), 10)) + (close / at_zero2nan(ts_mean(close, 20))))), 60), 60)
```

GP form: ts_std(ts_median(add(ts_delta(c_ma20, w10), add(sub(ts_min(ts_decay(ts_delta(beta60, w10), w_id(w60))), w60), ts_min(neg(c_ma20), w60))), add(signlog(ts_min(ts_delta(beta60, w10), w10)), c_ma20))), w_id

Alpha 14 — csv rank 8, seed 15

held-out IR **0.58** | daily IR **0.1015** (gate 0.07) | TVR **0.023** (limit 0.5) | net IR 1.52 | stocks 303 (min 160) | drawdown 1.7% | max corr 0.07

Paste into Trexsim:

```
alpha = ts_sum(ts_std(ts_max(ts_max(ts_delay((((close / at_zero2nan(ts_mean(close, 60))) * ts_skew(((close / at_zero2nan(ts_mean(close, 60))) * ((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1)))), 20))), 60), 10), 20), 60), 60)
```

GP form: ts_sum(ts_std(ts_max(ts_max(ts_delay(mul(c_ma60, ts_skew(mul(c_ma60, gap), w20))), w_id(w_id(w_id(w60)))), w_id(w10)), w_id(w20)), w_id(w_id(w_id(w60)))), w_id(w_id(w60)))

Alpha 15 — csv rank 19, seed 2

held-out IR **0.56** | daily IR **0.1592** (gate 0.07) | TVR **0.204** (limit 0.5) | net IR 1.64 | stocks 322 (min 160) | drawdown 1.0% | max corr 0.21

Paste into Trexsim:

```
alpha = (ts_mean((ts_corr_binary((((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20)))) + ((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1)))) - ts_delay(((close * volume) / at_zero2nan(ts_mean(close * volume, 60))), 20)), ts_corr_binary(ts_mean(ts_corr_binary(ret1, ret1_spx, 60), 60), ((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1))), 10), 10) - ts_zscore((close / at_zero2nan(ts_mean(close, 20))), 5)), 10) + ts_mean((ts_mean((ts_corr_binary(ts_std(ret1, 20), ts_zscore((close / at_zero2nan(ts_mean(close, 20))), 10), 10) - ts_zscore(((close * volume) / at_zero2nan(ts_mean(close * volume, 60))), 60)), 5) - (ts_std(ret1, 20) - ts_delay(((close * volume) / at_zero2nan(ts_mean(close * volume, 60))), 10))), 20))
```

GP form: add(ts_mean(sub(ts_corr(sub(add(neg(stoch), gap), ts_delay(dv_adv, w20))), ts_corr(ts_mean(beta60, w60), gap, w10), w10), ts_zscore(c_ma20, w_id(w5))), w10), ts_mean(sub(ts_mean(sub(ts_corr(vol20, ts_z

Alpha 16 — csv rank 28, seed 11

held-out IR **0.51** | daily IR **0.1292** (gate 0.07) | TVR **0.196** (limit 0.5) | net IR 1.20 | stocks 330 (min 160) | drawdown 1.3% | max corr 0.49

Paste into Trexsim:

```
alpha = ts_sum(-(ts_zscore(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))), 10)) - (((ts_median(cs_zscore((volume / at_zero2nan(ts_mean(volume, 20))), 20) - ts_mean(ret20, 10)) + ts_skew(((volume / at_zero2nan(ts_mean(volume, 20))) - ts_delay((volume / at_zero2nan(ts_mean(volume, 20))), 5)), 20)) - ts_mean(-(close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))), 10) - ts_median((ts_std(ret1, 20) / at_zero2nan(ret20)), 5)), 10))), 10)
```

GP form: ts_sum(sub(neg(ts_zscore(stoch, w_id(w_id(w_id(w10))))), sub(add(sub(ts_median(cs_zscore(v_ma20), w_id(w20))), ts_mean(ret20, w_id(w_id(w10)))), ts_skew(ts_delta(v_ma20, w5), w_id(w20))), ts_mean

Alpha 17 — csv rank 9, seed 15

held-out IR **0.51** | daily IR **0.0995** (gate 0.07) | TVR **0.033** (limit 0.5) | net IR 1.55 | stocks 307 (min 160) | drawdown 4.8% | max corr 0.27

Paste into Trexsim:

```
alpha = ts_sum(ts_std(ts_max(ts_delay((((ts_std(ret1, 20) - ts_delay(ts_std(ret1, 20), 10)) * ts_zscore(((close - open) / at_zero2nan(open)), 60)) * (ts_zscore((close / at_zero2nan(ts_mean(close, 60))), 5) * ((open - ts_delay(close, 1)) / at_zero2nan(ts_delay(close, 1))))), 60), 60), 5), 60)
```

GP form: ts_sum(ts_std(ts_max(ts_delay(mul(mul(ts_delta(vol20, w10), ts_zscore(co, w60)), mul(ts_zscore(c_ma60, w5), gap)), w60), w_id(w_id(w60))), w_id(w5)), w_id(w_id(w_id(w_id(w60)))))

Alpha 18 — csv rank 20, seed 26

held-out IR **0.47** | daily IR **0.0821** (gate 0.07) | TVR **0.055** (limit 0.5) | net IR 1.07 | stocks 319 (min 160) | drawdown 2.0% | max corr 0.15

Paste into Trexsim:

```
alpha = (ts_mean(ts_max(at_signlog(ts_delay(ts_corr_binary(ts_corr_binary(ret1, ret1_spx, 60), ((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))), 20), 10)), 20), 60) - at_signlog((close / at_zero2nan(ts_mean(close, 20)))))
```

GP form: sub(ts_mean(ts_max(signlog(ts_delay(ts_corr(beta60, stoch, w20), w_id(w10))), w_id(w20)), w60), signlog(c_ma20))

Alpha 19 — csv rank 15, seed 18

held-out IR **0.39** | daily IR **0.0915** (gate 0.07) | TVR **0.063** (limit 0.5) | net IR 1.17 | stocks 311 (min 160) | drawdown 1.1% | max corr 0.33

Paste into Trexsim:

```
alpha = (ts_min(ts_min(ts_min(ts_zscore(ts_min(ts_min(ts_zscore(ret5, 60), 20), 10), 60), 60), 5), 60) - cs_zscore(ts_mean_exp(cs_zscore((ts_mean_exp(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))), 5, 0.5) + (cs_rank((cs_rank((cs_rank(ts_mean(close * volume, 60)) - 1)) - 1)) - 1))), 5, 0.5)))
```

GP form: sub(ts_min(ts_min(ts_min(ts_zscore(ts_min(ts_min(ts_zscore(ret5, w60), w20), w10), w60), w60), w5), w60), cs_zscore(ts_decay(cs_zscore(add(ts_decay(stoch, w5), cs_rank(cs_rank(size)))), w5)))

Alpha 20 — csv rank 23, seed 10

held-out IR **0.13** | daily IR **0.0830** (gate 0.07) | TVR **0.073** (limit 0.5) | net IR 0.99 | stocks 316 (min 160) | drawdown 1.6% | max corr 0.13

Paste into Trexsim:

```
alpha = ts_corr_binary(ts_max((ts_mean(ts_std(ret1, 20), 60) / at_zero2nan(ts_std(ret1, 20))), 60), ts_mean_exp(ts_mean_exp(ts_std(ret1, 20), 60, 0.5), 60, 0.5), 60)
```

GP form: ts_corr(ts_max(div(ts_mean(vol20, w60), vol20), w_id(w_id(w60))), ts_decay(ts_decay(vol20, w60), w60), w60)

Alpha 21 — csv rank 13, seed 5

held-out IR **0.04** | daily IR **0.1077** (gate 0.07) | TVR **0.077** (limit 0.5) | net IR 1.38 | stocks 323 (min 160) | drawdown 2.1% | max corr 0.10

Paste into Trexsim:

```
alpha = ts_mean_exp(((ret5 + ts_corr_binary(ts_median(-(ret20), 60), (close / at_zero2nan(ts_mean(close, 20))), 60)) * -(((close - ts_min(low, 20)) / at_zero2nan(ts_max(high, 20) - ts_min(low, 20))))) , 20, 0.5)
```

GP form: ts_decay(mul(add(ret5, ts_corr(ts_median(neg(ret20), w_id(w60)), c_ma20, w60)), neg(stoch)), w_id(w_id(w_id(w20))))